Enrollment No: _____

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B. Tech Mid Semester Exam

Semester: 6th SEM | **Date: (17/02/2026)**
Subject Code: 303105353 | **Time: (1hr: 30min)**
Subject Name: Machine Learning | **Total Marks: 40**

Sr. No.	Questions:	Marks	Co	Blooms Taxonomy
Q.1	**(A) One-line Questions:** (i) What are the primary differences between the Linear and Logistic regression models? (ii) What is a confusion matrix? (iii) What is the purpose of a training dataset and test dataset? (iv) What is overfitting and underfitting? (v) What is a missing value?	05	CO-1 and CO-2	1
	(B) Fill in the blanks: 1. In linear regression, the goal is to fit a line that minimizes the _____ between predicted and actual values, often using the method of _____. 2. The process of replacing missing values in a dataset is part of _____, which ensures the data is suitable for analysis. 3. The _____ method is a common statistical approach to detect outliers based on their distance from the mean in terms of standard deviations. 4. In machine learning, _____ spaces represent the set of all hypotheses consistent with the training data. 5. Normalization is a data preprocessing technique where the values of features are scaled to fit within a specified range, often between _____ and _____.	05	CO-1 and CO-2	1 & 2
Q.2	**Attempt any four (Short Questions):**	12		
	(1) What are the main learning paradigms in machine learning?		CO-1	1
	(2) Define the concept of version spaces in Hypothesis evaluation.		CO-1	1
	(3) Evaluate the effectiveness of Hyperparameter tuning in improving model accuracy.		CO-1	4
	(4) Explain the Z- Score analysis to identify outliers?		CO-1	3
	(5) Describe the role of "Probability" in Machine Learning?		CO-1	2

Q.3	**Attempt any two questions:**	08		
	(1) Analyze the differences between multi-class and multi-label classification, providing examples of scenarios where each is applicable.		CO-2	4
	(2) What is PAC learning, and how does it help in evaluating the efficiency of a learning algorithm?		CO-2	2
	(3) Given the following training data for K-NN Classification: Point | X | Y | Class A | 1 | 2 | Red B | 2 | 3 | Red C | 3 | 3 | Blue D | 6 | 5 | Blue A new point **P (2, 2)** is given. Classify P using **K = 3** and **Euclidean distance**.		CO-1	3
Q.4	(A) A dataset contains values: [10,12,15,18,100]. (i) Calculate the mean and standard deviation. (ii) Compute the Z-Score for each value.	05	CO-1	2
	(B) What do you mean by Data Visualization? Explain any four types of result Visualization Techniques.	05	CO-1	1
	OR			
	(B) Describe the steps involved in preprocessing a dataset, including handling missing values, normalization, and adapting the data to fit a chosen algorithm.	05	CO-1	2